$$A\underline{x} = \lambda \underline{x}$$

<u>Self-adjoint</u>

$$A = \underset{\sim}{U} \Lambda U^H \qquad U: \text{unitary}$$

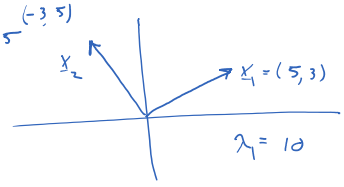
$$\Lambda = \begin{bmatrix} \lambda_1 & & & \\ & \lambda_m & & \\ & & \ddots & \\ & & & \lambda_m \end{bmatrix}$$

$$= \sum_{i=1}^{m} \lambda_i \underline{y}_i \underline{y}_i^H$$

$$\underbrace{[\,|\,][\,-\,]}$$

want R:

$$\underline{x}_2 = \frac{1}{\sqrt{34}}\begin{bmatrix} -3 \\ 5 \end{bmatrix} \qquad \lambda_2 = 5$$



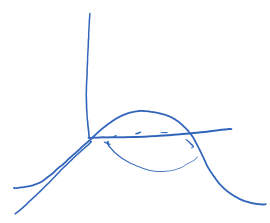
$$\underline{x}_1 = \frac{1}{\sqrt{34}}\begin{bmatrix} 5 \\ 3 \end{bmatrix}$$

$$R = (10)\left(\frac{1}{34}\right)\begin{bmatrix} 5 \\ 3 \end{bmatrix}\begin{bmatrix} 5 & 3 \end{bmatrix} + 5\left(\frac{1}{34}\right)\begin{bmatrix} -3 \\ 5 \end{bmatrix}\begin{bmatrix} -3 & 5 \end{bmatrix}$$

$$X \sim N(\mu, R)$$

$$f(\underline{x}) = (k) \exp\left[-\frac{1}{2}(\underline{x}-\underline{\mu})^T R^{-1} (\underline{x}-\underline{\mu})\right]$$

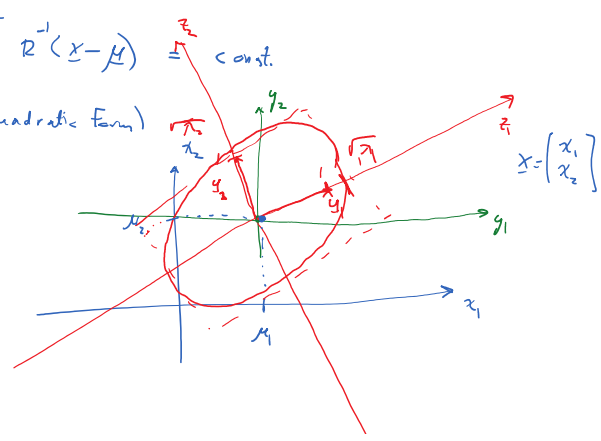
want plot $\underline{x}$ s.t. $f(\underline{x}) =$ constant.
(contour plot)



$f(\underline{x}) =$ cons.

$$\log f(x) \longrightarrow (\underline{x}-\underline{\mu})^T R^{-1} (\underline{x}-\underline{\mu}) = \text{const.}$$

(Quadratic Form)



$$\underline{y} = \underline{x} - \underline{\mu}$$

$$\underline{y}^T R^{-1} \underline{y}$$

$$R = U \Lambda U^T$$

$$R = U \Lambda U^T$$

$$U = [\, y_1 \;\; y_2 \,] \qquad U^T U = I$$

$$\Lambda = \begin{bmatrix} \lambda_1 & \\ & \lambda_2 \end{bmatrix}$$

$$z = U^T y \qquad y = U z$$

Suppose $z = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$

$y = Uz = y_1$

$z = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

$y = Uz = y_2$

$$R^{-1} = U \Lambda^{-1} U^T$$

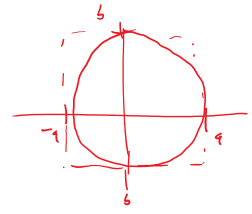
$$y^T R^{-1} y = z^T \cancel{U^T} \cancel{U} \Lambda^{-1} \cancel{U^T} \cancel{U} z$$

$$= z^T \Lambda^{-1} z$$

$$= [\, z_1 \;\; z_2 \,] \begin{bmatrix} 1/\lambda_1 & \\ & 1/\lambda_2 \end{bmatrix} \begin{bmatrix} z_1 \\ z_2 \end{bmatrix}$$

$$= \frac{z_1^2}{\lambda_1} + \frac{z_2^2}{\lambda_2} = \text{const.}$$

$$\frac{x^2}{4^2} + \frac{y^2}{3^2} = 1$$



<u>Rayleigh Quotient</u>

$$R_A(\underline{x}) = \frac{\underline{x}^H A \underline{x}}{\underline{x}^H \underline{x}} \qquad A: \text{ self adjoint}$$

$$\max_{\underline{x} \neq 0} R_A(\underline{x})$$

$$\max_{\|\underline{x}\|_2 = 1} R_A(\underline{x}) \qquad \begin{array}{ll} \max & \underline{x}^H A \underline{x} \\ \text{s.t.} & \underline{x}^H \underline{x} = 1 \end{array}$$

$$A \underline{x} = \lambda \underline{x}$$

$$(A + \sigma^2 I)\, y = \mu y$$

$$A y + \sigma^2 y = \mu y$$

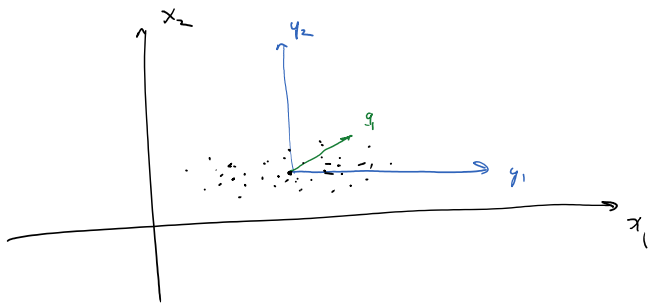
$$\boxed{A y = (\mu - \sigma^2)\, y}$$

$y = \underline{x}$ (eigenvectors of $A + \sigma^2 I$ are the same as for A)

$$\mu - \sigma^2 = \lambda$$

$$\boxed{\mu = \lambda + \sigma^2}$$

Principal Components



Data: $\underline{x}_1, \underline{x}_2, \ldots, \underline{x}_N$, $\underline{x}_i \in \mathbb{R}^d$

Mean: $\frac{1}{N}\sum_{i=1}^{N} \underline{x}_i = \bar{\underline{x}}$

$$\underline{y}_i = \underline{x}_i - \bar{\underline{x}}$$

Direction vectors: $\underline{q}_1$: $\|q_1\| = 1$ $\quad \underline{q}_1^T \underline{q}_1 = 1$

component of $\underline{y}_i$ in the $\underline{q}_1$ direction is

$$\langle y_i, q_1 \rangle = q_1^T y_i$$

Form $J = \frac{1}{N}\sum_{i=1}^{N} (\underline{q}_1^T \underline{y}_i)^2$

Goal: choose q_1 to make J as large as possible.

Let $S = \underbrace{\frac{1}{N}\sum_{i=1}^{N} \underline{y}_i \underline{y}_i^T}$ (scatter matrix)

$$J = \frac{1}{N}\sum_{i=1}^{N} q_1^T y_i y_i^T q_1$$

$$= q_1^T \frac{1}{N}\left[\sum_{i=1}^{N} y_i y_i^T\right] q_1$$

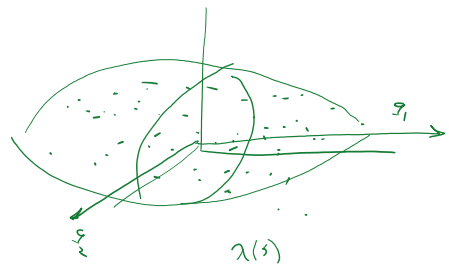
$$= q_1^T S q_1$$

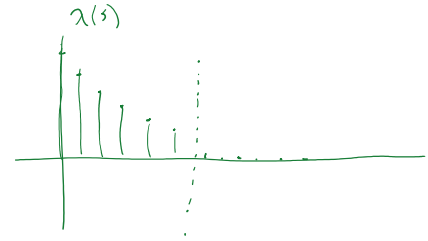
$$\left[\begin{array}{ll} \max & q_1^T S q_1 \\ \text{s.t.} & q_1^T q_1 = 1 \end{array}\right.$$

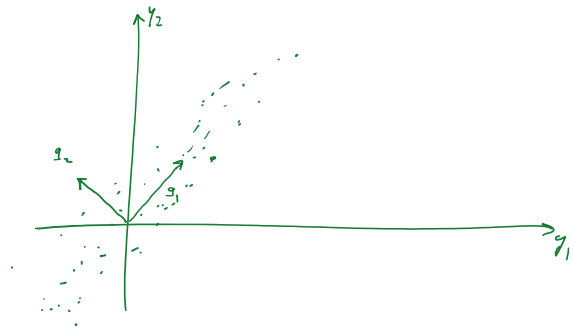
$$\max_{q_2} \quad q_2^T S q_2$$

s.t. $\quad q_2^T q_2 = 1$

$\quad q_2^T q_1 = 0$





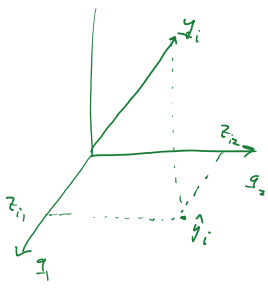


$$z_{i1} = q_1^T y_i$$

$$z_{i2} = q_2^T y_i$$

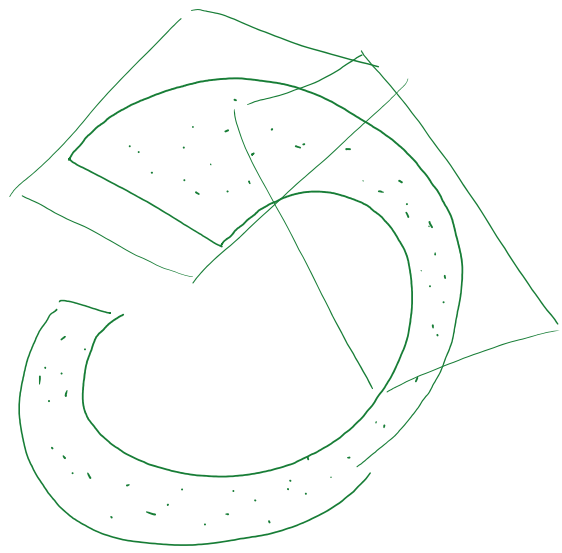
$$\vdots$$

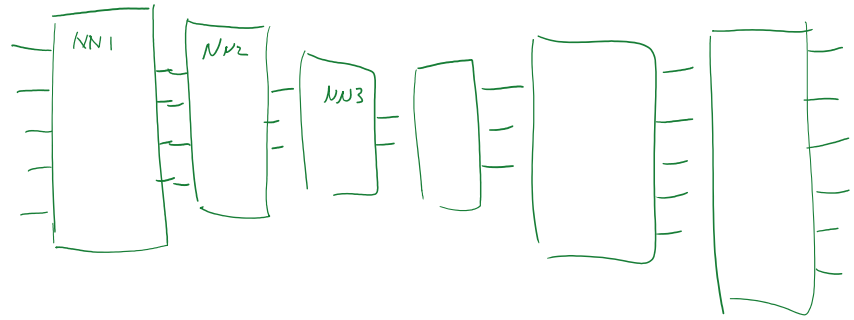
$$z_{ip} = q_p^T y_i$$



$$\begin{bmatrix} z_{i1} \\ z_{i2} \\ \vdots \\ z_{ip} \end{bmatrix}$$

$$\hat{y}_i = z_{i1} q_1 + z_{i2} q_2 + \cdots + z_{ip} q_p$$





$$y_i \, y_i^T$$

$$\begin{bmatrix} y_{i1} \\ y_{i2} \\ \vdots \\ y_{i200} \end{bmatrix} \begin{bmatrix} y_{i1} & y_{i2} & \cdots & y_{i200} \end{bmatrix}$$

Signal subspace Techniques

$$x[t] = A \sin(\omega_0 t)$$

$$\downarrow$$

$$x[t+2] + a_1 x[t+1] + a_2 x[t] = 0$$

Signal model:

$$x[t] = \sum_{i=1}^{\overset{\text{p modes}}{P}} \underbrace{a_i}_{\text{amplitude}} e^{j(2\pi \overset{f_i = \text{freq.}}{f_i} t + \underbrace{\phi_i}_{\text{phase}})}$$

$$f_i \in [-0.3, 0.5]$$

ϕ_i : phase
random
independent $\sim \mathcal{U}(0, 2\pi)$

Autocorrelation:

$$r_{xx}[k] = E[x[t] \, x[t-k]^*]$$

$$= E\left[\left(\sum_{i=1}^{P} a_i e^{j(2\pi f_i t + \phi_i)}\right)\left(\sum_{l=1}^{P} a_l^* e^{-j(2\pi f_l (t-k) + \phi_l)}\right)\right]$$

$$= \sum_{i=1}^{P} \sum_{l=1}^{P} a_i a_l^* e^{j(2\pi f_i t)} e^{-j2\pi f_l (t-k)} E\left[e^{j\phi_i} e^{-j\phi_l}\right]$$

$$\underbrace{E\left[e^{j(\phi_i - \phi_l)}\right]}_{= \begin{cases} 1 & \text{if } i = l \\ 0 & \text{if } i \neq l \end{cases}}$$

$$= \sum_{i=1}^{P} |a_i|^2 e^{j2\pi f_i k}$$

Form $\underline{x}[t] = \begin{bmatrix} x[t] \\ x[t+1] \\ \vdots \\ x[t+M-1] \end{bmatrix}$ ($M \times 1$ vector)

$$R_{xx} = E\left[\underline{x}[t]\ \underline{x}[t]^H\right] = E\left(\begin{bmatrix} x[t] \\ x[t+1] \\ \vdots \\ x[t+M-1] \end{bmatrix}\left[x[t]^* \quad x[t+1]^* \cdots \quad x[t+M-1]^*\right]\right)$$

$$= \begin{bmatrix} r_{xx}[0] & r_{xx}[-1] & \cdots & r_{xx}[-(M-1)] \\ r_{xx}[1] & r_{xx}[0] & \cdots & r_{xx}[-(M-2)] \\ \vdots & & & \\ r_{xx}[M-1] & r_{xx}[M-2] & \cdots & r_{xx}[0] \end{bmatrix}$$

$$R_{xx}^H = R_{xx}$$

$$R_{xx} = \sum_{i=1}^{p} |q_i|^2\ \underline{s}_i\ \underline{s}_i^H, \qquad \underline{s}_i = \begin{bmatrix} 1 \\ e^{j2\pi f_i} \\ e^{j2\pi 2 f_i} \\ \vdots \\ e^{j2\pi(M-1)f_i} \end{bmatrix}$$

$$S = \begin{bmatrix} \underline{s}_1 & \underline{s}_2 & \cdots & \underline{s}_p \end{bmatrix}$$

$$\underbrace{R_{xx} = S P S^H}$$

$$P = \begin{bmatrix} |q_1|^2 & & & \\ & |q_2|^2 & & \\ & & \ddots & \\ & & & |q_p|^2 \end{bmatrix}$$

If we do the eigen d

$$R_{xx} = U \Lambda U^H \qquad U = [\underline{u}_1\ \underline{u}_2 \cdots \underline{u}_M]$$

$$\Lambda = \begin{bmatrix} \lambda_1 & & & \\ & \lambda_2 & & \\ & & \ddots & \\ & & & \lambda_M \end{bmatrix}$$

Lemma: If $M \geq p$

$$\text{span}(\underline{u}_1, \underline{u}_2, \ldots, \underline{u}_p) = \underline{\text{span}(\underline{s}_1, \underline{s}_2, \ldots, \underline{s}_p)}$$

Define $\text{span}(\underline{u}_1, \underline{u}_2, \ldots, \underline{u}_p)$ = "signal space"

Let $y[t] = x[t] + w[t]$

$w[t]$: zero mean white

$E[w[t]\, w^*[t]] = \sigma_w^2$

$w[t]$ is uncorrelated with the $x[t]$.

$$r_{yy}[k] = E[y[t]\, y^*[t-k]]$$
$$= E[(x[t] + w[t])(x[t-k]^* + w[t-k]^*)]$$
$$= r_{xx}[k] + \sigma_w^2\, \delta[k]$$

$$\delta[k] = \begin{cases} 1 & \text{if } k=0 \\ 0 & \text{if } k \neq 0 \end{cases}$$

$$\underline{y}[t] = \begin{bmatrix} y[t] \\ y[t+1] \\ \vdots \\ y[t+M-1] \end{bmatrix}$$

$$R_{yy} = E[\underline{y}[t]\, \underline{y}[t]^H]$$
$$= \underbrace{\underbrace{R_{xx}}_{\text{rank } p} + \sigma_w^2 I}_{\substack{\text{full rank} \\ \text{rank} = M}}$$

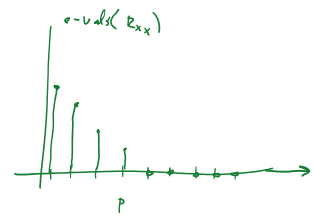
Assume $M \geq p$

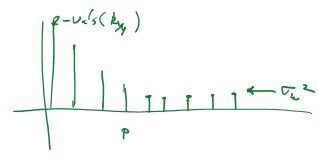
If $R_{xx}\, \underline{u} = \lambda\, \underline{u}$

$\underline{u}$ is e-vect of R_{xx}

λ is e-val of R_{xx}

$$R_{yy}\, \underline{u} = (R_{xx} + \sigma_w^2 I)\underline{u}$$
$$= R_{xx}\, \underline{u} + \sigma_w^2\, \underline{u}$$
$$= \lambda\, \underline{u} + \sigma_w^2 \underline{u} = (\lambda + \sigma_w^2)\underline{u}$$





Pisarenko Harmonic Decomposition

(PHD)

Let $M = p+1$

$\underline{u}_M \perp \mathcal{S}$

$\underbrace{\underline{u}_1\ \underline{u}_2 \cdots \underline{u}_p}_{\mathcal{S}}\ \underbrace{\underline{u}_{p+1}\ \underline{u}_{p+2} \cdots \underline{u}_M}_{N}$

$N = \text{span}(\underline{u}_{p+1}, \underline{u}_{p+2}, \ldots \underline{u}_M)$

$\mathcal{S} \perp N$

$\underline{u}_M \perp \underline{s}_1 \quad \underline{u}_M \perp \underline{s}_2, \ldots \quad \underline{u}_M \perp \underline{s}_p$

$\underline{u}_M \perp \underline{s}_1 \quad \underline{u}_M \perp \underline{s}_2, \ldots \quad \underline{u}_M \perp \underline{s}_p$

$\underline{s}_i^H\, \underline{u}_M = 0 \qquad i = 1, 2, \ldots, p$

$$\underline{s}_i = \begin{bmatrix} 1 \\ e^{j2\pi f_i} \\ e^{j2\pi 2 f_i} \\ \vdots \\ e^{j2\pi(M-1) f_i} \end{bmatrix}$$

$$\sum_{k=0}^{M-1} u_{Mk}\, e^{-j2\pi f_i k} = 0$$

Define $\gamma = e^{-j2\pi f_i}$

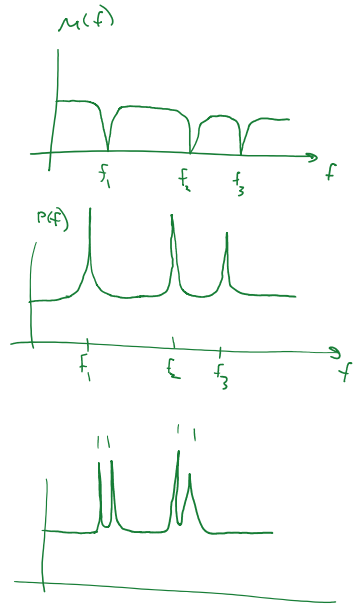
$$\sum_{k=0}^{M-1} u_{mk}\gamma^k = 0 \qquad u_{m0} + u_{m1}\gamma + u_{m2}\gamma^2 + \cdots + u_{mp}\gamma^p = 0$$

MUSIC — multiple signal classification

$$\underline{s}(f) = \begin{bmatrix} 1 \\ e^{i2\pi f} \\ e^{i2\pi 2f} \\ \vdots \\ e^{i2\pi(M-1)f} \end{bmatrix}$$

$$M(f) = \sum_{k=p+1}^{M} \left| s(f)^H u_k \right|^2$$

$$P(f) = \frac{1}{M(f)}$$



"Music spectrum"

Singular Value Decomp. (SVD)

A: square, self-adjoint $\qquad A = U \Lambda U^H$

U: unitary — eigenvectors $U^H U = I$

Λ: diagonal — eigenvalues.

More generally, let A be general.

- Not necessarily square
- Not necessarily self-adjoint.

$A: m \times n$

$$A = U \Sigma V^H$$

where: U: $m \times m$ unitary

V: $n \times n$ unitary

Σ: $m \times n$ "diagonal."

$p = \min(m, n)$

$$\Sigma = \begin{bmatrix} \sigma_1 & & & \\ & \sigma_2 & & \\ & & \ddots & \\ & & & \sigma_p \\ & & & \end{bmatrix}$$

$$\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_p \geq 0$$

$$A = U \Lambda U^H$$

$$AA^H = (U\Sigma\cancel{V^H})\cancel{V}\Sigma^T U^H$$
$$= U\Sigma^2 U^H$$

$$A^H A = V\Sigma^T\cancel{U^H}\cancel{U}\Sigma V^H$$
$$= V\Sigma^2 V^H$$